

Elephant Mortality Analysis

Raigarh & Dharamjaigarh Forest Divisions · District Raigarh, Chhattisgarh

Based on official divisional death registers, 2020–2026 (n = 51)

Prepared for: Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · 5–6 June 2026, Raigarh · Technical support: WII Dehradun

1. Executive Summary

This report analyses **51 elephant deaths** recorded by two adjacent forest divisions of Raigarh district — **Dharamjaigarh (31, calendar years 2020–2026)** and **Raigarh (20, financial years 2021-22 to 2026-27)**. Two causes dominate: **electrocution (21, 41%)** and **drowning in water bodies (11, 22%)**, together accounting for 63% of all deaths. Of the total, **22 (43%) were unnatural** and **29 natural**; **25 (49%) were calves or juveniles under five years**. The two divisions show distinct mortality signatures — Dharamjaigarh loses prime adults to illegal-wire electrocution, while Raigarh loses calves to drowning in dams and ponds.

Metric	Dharamjaigarh	Raigarh	Combined
Total deaths	31	20	51
Electrocution	13	8	21
Drowning / water body	4	7	11
Unnatural (all causes)	13	9	22
Calves/juveniles (<5 yr)	10	15	25
Legal action (accused booked)	12	0	12

2. Data Sources & Method

Two official registers of the Chhattisgarh Forest Department were transcribed and translated: (a) 'Elephant Death 2020 to 2026', Dharamjaigarh Forest Division; and (b) 'Last Five Years Elephant Death', Raigarh Forest Division. Every dead elephant is one record; multi-death incidents were split into separate rows.

Causes stated in the registers (per post-mortem reports) were mapped to standardised categories for comparison. Dharamjaigarh records run by calendar year; Raigarh by financial year — for temporal charts the calendar year of the incident date is used. Ages given as ranges/months were converted to a mid-point in years for age-class assignment. A small number of calf sexes were not stated in the source and are marked 'Unknown'.

3. Key Findings

3.1 Electrocution is the single largest killer.

21 of 51 deaths (41%) were caused by electrocution — 13 in Dharamjaigarh and 8 in Raigarh. It accounts for 21 of the 22 'unnatural' deaths (the 22nd being a train hit). Most Dharamjaigarh cases involved illegal live wires strung around crop fields; 11 of 21 electrocution victims were prime adults (>15 yr), the reproductively most valuable animals.

3.2 Drowning in water bodies is the second-largest cause and an emerging crisis.

11 deaths (22%) were drownings in dams, ponds and marshes — 7 in Raigarh and 4 in Dharamjaigarh. Almost all victims were calves. Department-built ponds and specific dams (Panikhet, Rabo, Ghoghra, Saraimuda, Amabada, Kera Jharia) recur in the records, indicating a man-made infrastructure hazard concentrated in the Gharghoda and Chhal ranges.

3.3 Calf and juvenile mortality dominates.

25 of 51 deaths (49%) were calves/juveniles under 5 years. Raigarh's mortality is overwhelmingly of calves (drowning, weakness, brain injury, heat stroke), whereas Dharamjaigarh loses more prime adults to electrocution — two distinct mortality signatures in adjacent divisions.

3.4 A 2026 infectious-disease cluster in Dharamjaigarh calves warrants surveillance.

Within weeks in April–May 2026, three Dharamjaigarh calves (1–6 months) died of acute serohaemorrhagic pneumonia, acute bacterial hepatitis, and bacterial septicaemia respectively. Such a temporally clustered set of fatal infections in young calves should trigger active EEHV / bacterial disease surveillance, histopathology and PCR on future calf deaths.

3.5 Enforcement is recorded only for electrocution, and only in one division.

Accused were booked in 12 cases — every one an electrocution in Dharamjaigarh. Raigarh's register does not capture an action column, and the single accidental 11 kV-line electrocution had no accused. This points to (a) genuine offences behind illegal-wire deaths and (b) an inconsistency in how the two divisions record enforcement.

3.6 Post-mortem coverage is complete; laboratory backing is improving.

Every death carries a post-mortem reference. From 2024 onward an increasing share also carries a laboratory report ('PM & Lab'), a positive trend that this workshop's necropsy-and-sampling training directly reinforces.

3.7 Mortality peaked in 2023 (Dharamjaigarh) and 2024–25 (Raigarh).

Dharamjaigarh's worst year was 2023 with 8 deaths, six of them electrocution. Raigarh's toll is concentrated in 2024–2026, driven by the calf-drowning cluster. Lightning (2 deaths, 2022) and inter-elephant fight (1) round out the natural causes.

4. Visual Analysis

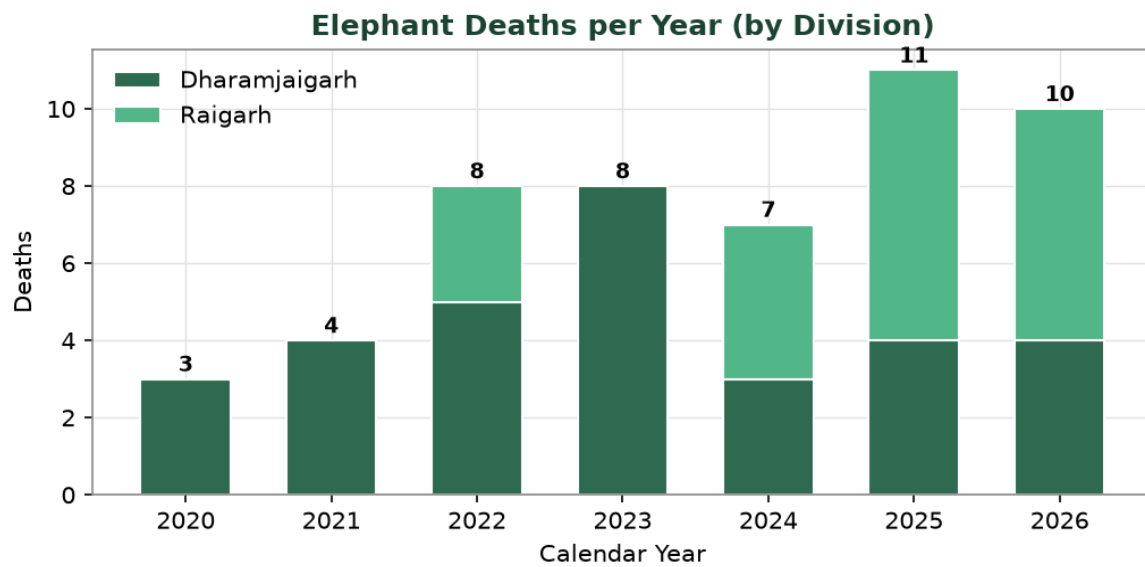


Fig 1 — Elephant deaths per year, by division

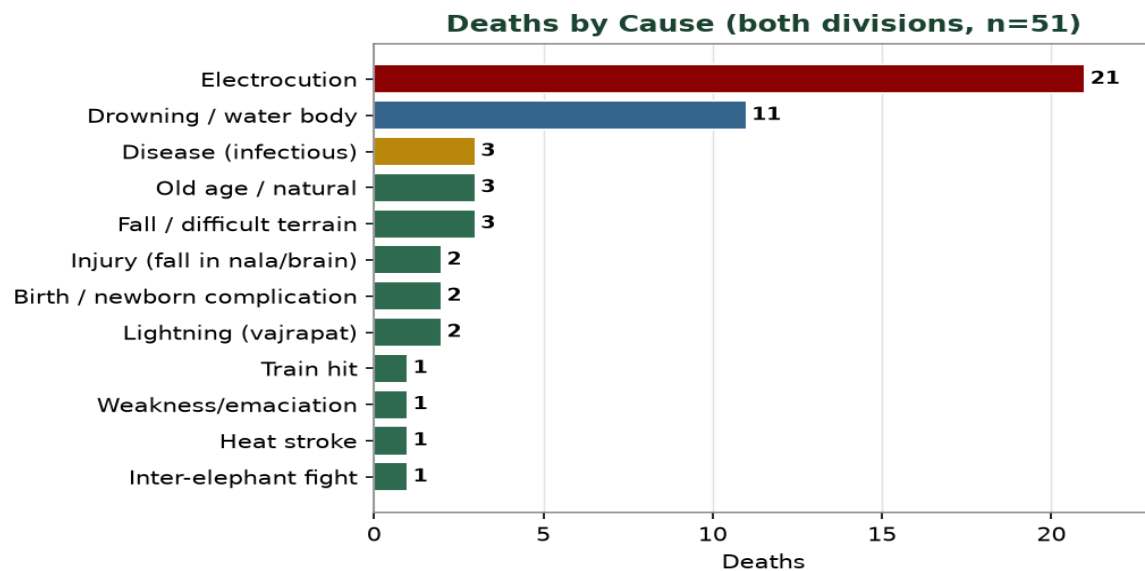


Fig 2 — Deaths by cause (both divisions, n=51)

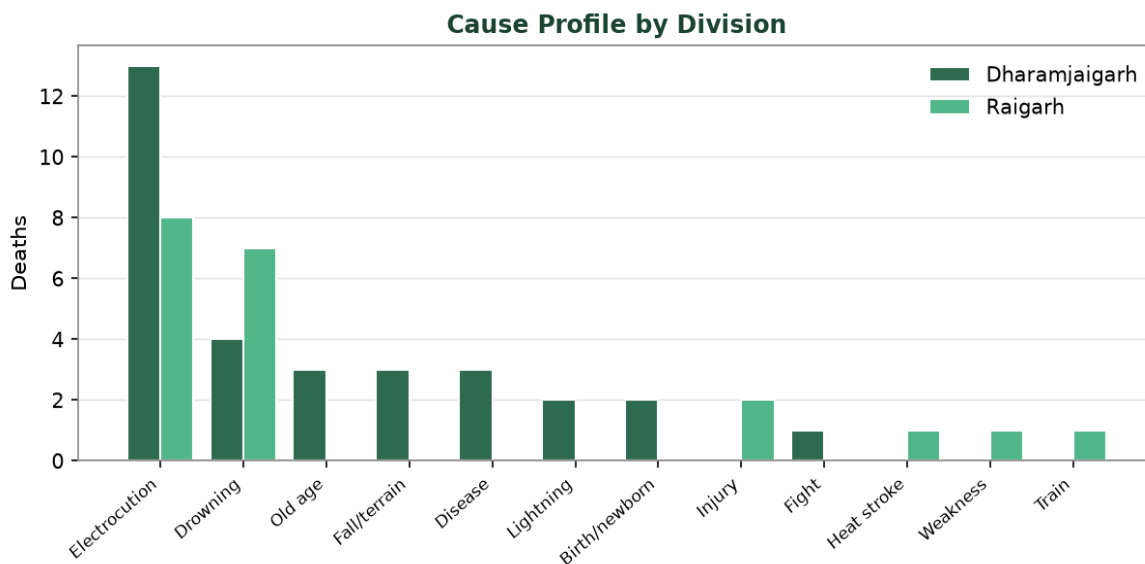


Fig 3 — Cause profile compared across the two divisions

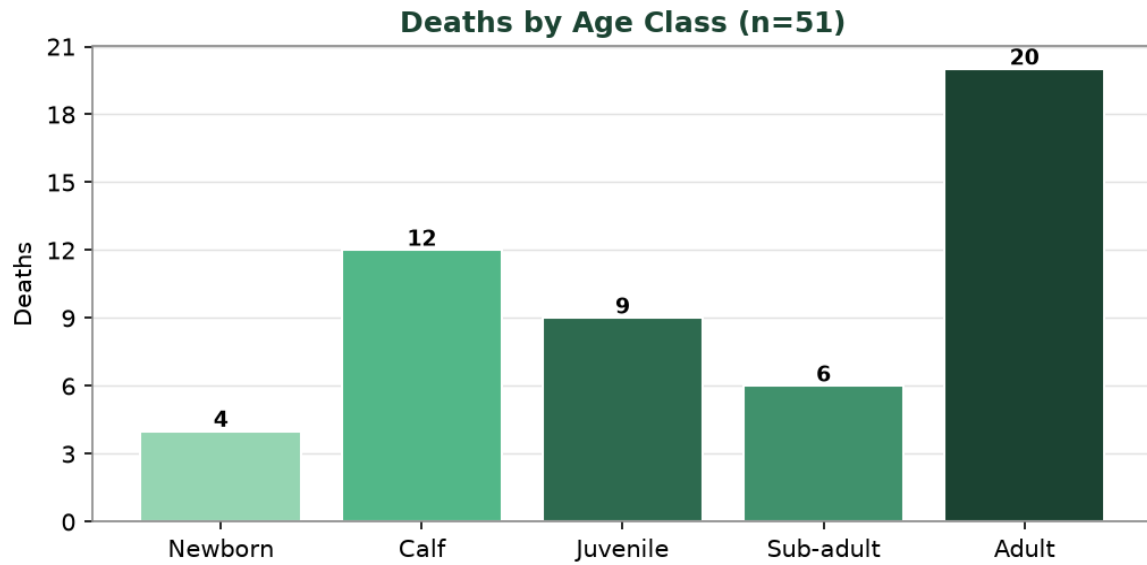


Fig 4 — Deaths by age class

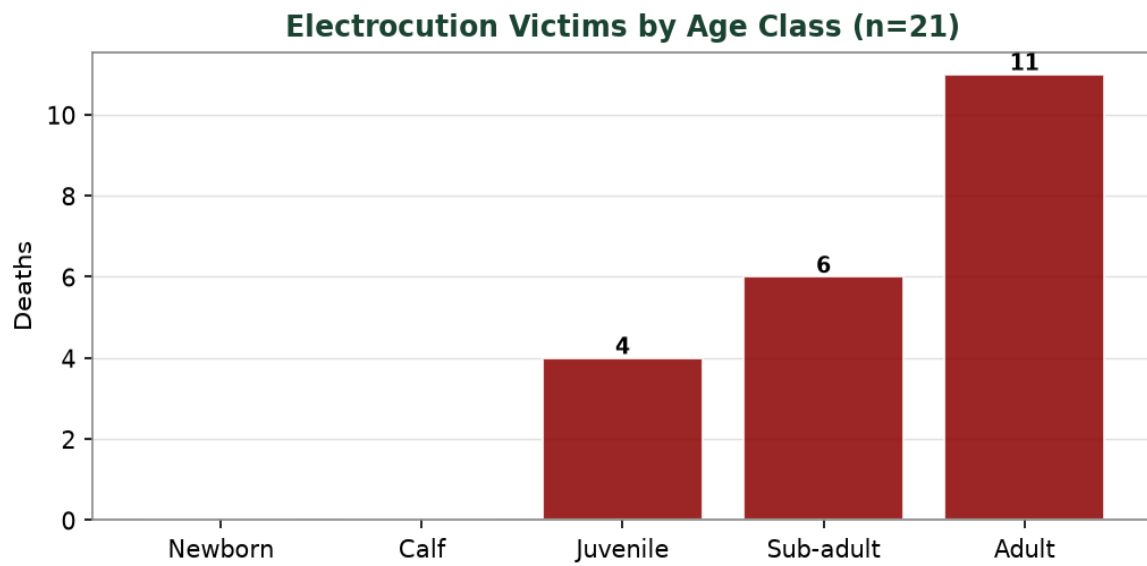


Fig 5 — Electrocution victims by age class

Natural vs Unnatural Deaths

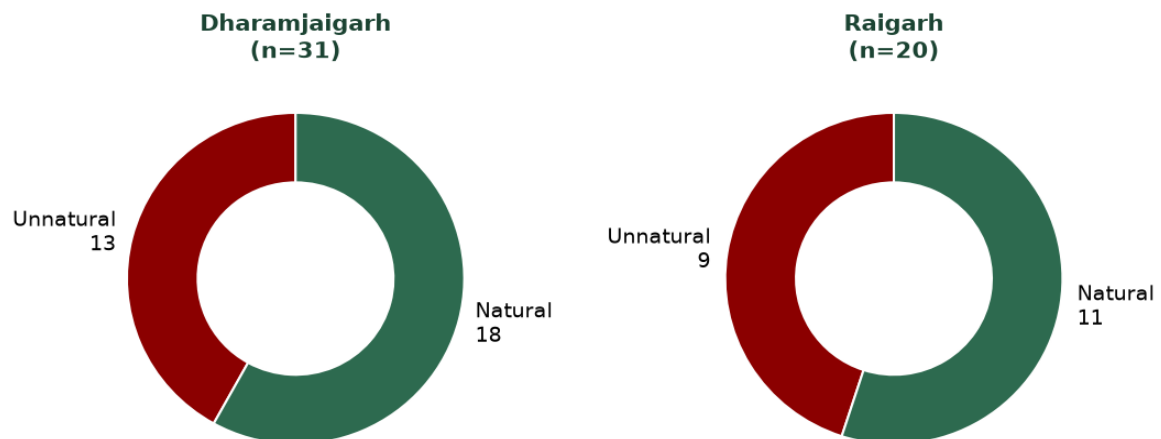


Fig 6 — Natural vs unnatural deaths per division

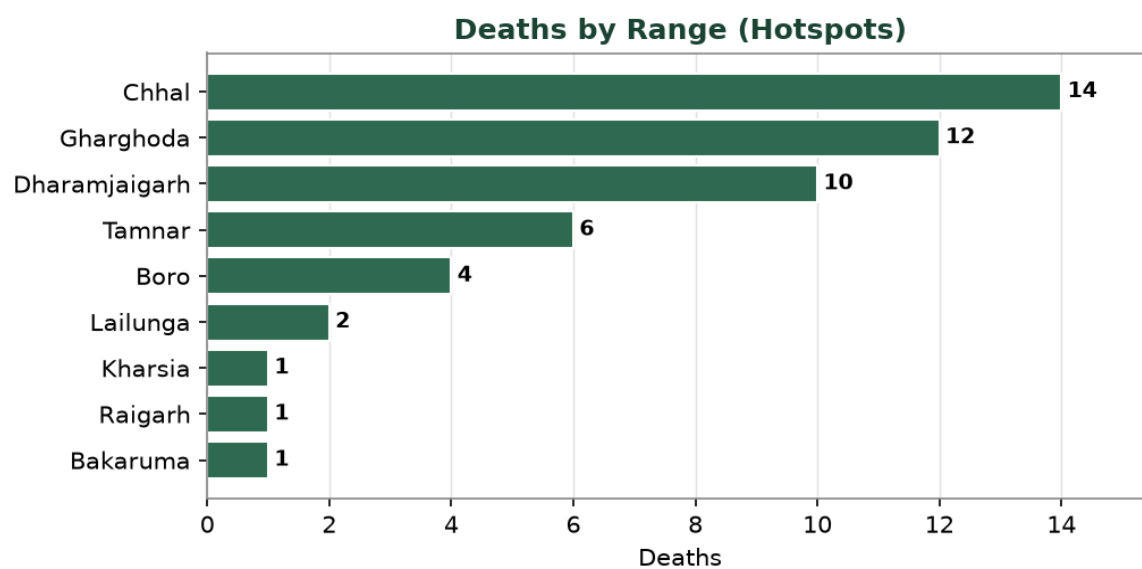


Fig 7 — Range-wise hotspots

5. Data Tables

5.1 Deaths by year & division

Year	Dharamjaigarh	Raigarh	Total
2020	3	0	3
2021	4	0	4
2022	5	3	8
2023	8	0	8
2024	3	4	7
2025	4	7	11
2026	4	6	10
TOTAL	31	20	51

5.2 Deaths by cause

Cause group	Dharamjaigarh	Raigarh	Total	%
Electrocution	13	8	21	41%
Drowning / water body	4	7	11	22%
Fall / difficult terrain	3	0	3	6%
Old age / natural	3	0	3	6%
Disease (infectious)	3	0	3	6%
Lightning (vajrapat)	2	0	2	4%
Birth / newborn complication	2	0	2	4%
Injury (fall in nala/brain)	0	2	2	4%
Inter-elephant fight	1	0	1	2%
Heat stroke	0	1	1	2%
Weakness/emaciation	0	1	1	2%
Train hit	0	1	1	2%

6. Recommendations

6.1 Electrocution — joint enforcement & line safety

- Constitute a standing Forest–Electricity Board task force for Chhal, Dharamjaigarh, Boro, Bakaruma (DJH) and Tamnar, Gharghoda (RGH) ranges.
- Audit, insulate/raise or underground sagging 11 kV lines in elephant corridors; enforce statutory ground clearance.
- Detect and dismantle illegal hooked live-wire fences; replace with legal pulsed solar energisers.
- Register FIRs under WPA 1972 (Sec 51) + Electricity Act (Sec 135/161); invoke strict liability of utilities.
- Standardise the enforcement/action column in BOTH divisions' registers.

6.2 Drowning — water-body engineering

- Barricade or fence steep-sided department ponds/dams implicated in deaths (Panikhet, Rabo, Ghoghra, Saraimuda, Amabada, Kera Jharia).
- Provide gentle-slope exit ramps / graded banks so calves can climb out.
- Prioritise Gharghoda and Chhal ranges, where calf drownings cluster.
- Audit all new water structures in elephant-movement zones for calf-safety before commissioning.

6.3 Calf health & disease surveillance

- Activate EEHV / bacterial disease surveillance for calf deaths; collect fresh + fixed tissue for PCR and histopathology.
- Investigate the April–May 2026 Dharamjaigarh calf cluster as a potential outbreak.
- Train field vets in calf-specific necropsy and sampling (this workshop).

6.4 Data & monitoring

- Merge both divisions onto one standardised register (identical fields, fixed cause categories, geo-tag, action column).
- Maintain the accompanying Excel database and update after every mortality.
- Produce an annual division-level mortality analysis for management review.

PRIORITY: Electrocution and calf-drowning together cause two-thirds of all deaths and are almost entirely PREVENTABLE. A Forest–Electricity Board task force on illegal wires and 11 kV sag, plus barricading/ramping of department water bodies in Gharghoda and Chhal ranges, would address the two largest killers directly.

7. Annexure — Full Line-List (n=51)

#	Div	Date	Range	Cause	Nat	Sex	Age	Age class	Act
1	DHAR	2020-06-16	Dharamjaigarh	Electrocution	Unnat	M	12	Sub-adult	Y
2	DHAR	2020-06-18	Chhal	Electrocution	Unnat	M	37	Adult	Y
3	DHAR	2020-09-23	Dharamjaigarh	Electrocution	Unnat	M	30	Adult	Y
4	DHAR	2021-03-29	Boro	Fall	Natur	M	18	Adult	—
5	DHAR	2021-06-22	Chhal	Electrocution	Unnat	F	15	Sub-adult	Y
6	DHAR	2021-08-14	Boro	Old age	Natur	F	52	Adult	—
7	DHAR	2021-08-22	Dharamjaigarh	Electrocution	Unnat	M	16	Adult	Y
8	DHAR	2022-02-04	Lailunga	Fall	Natur	F	42	Adult	—
9	DHAR	2022-07-31	Chhal	Lightning	Natur	M	41	Adult	—
10	DHAR	2022-09-26	Bakaruma	Lightning	Natur	F	45	Adult	—
11	DHAR	2022-10-14	Chhal	Electrocution	Unnat	M	20	Adult	Y
12	DHAR	2022-11-22	Dharamjaigarh	Fall	Natur	M	62	Adult	—
13	DHAR	2023-01-10	Dharamjaigarh	Electrocution	Unnat	M	75	Adult	Y
14	DHAR	2023-02-04	Dharamjaigarh	Old age	Natur	F	75	Adult	—
15	DHAR	2023-04-17	Chhal	Electrocution	Unnat	M	5	Sub-adult	Y
16	DHAR	2023-06-15	Chhal	Inter-elephant fight	Natur	M	1	Juvenile	—
17	DHAR	2023-09-10	Dharamjaigarh	Electrocution	Unnat	M	45	Adult	Y
18	DHAR	2023-10-11	Dharamjaigarh	Electrocution	Unnat	M	50	Adult	—
19	DHAR	2023-10-23	Chhal	Electrocution	Unnat	F	42	Adult	Y
20	DHAR	2023-12-20	Boro	Electrocution	Unnat	M	15	Sub-adult	Y
21	DHAR	2024-09-03	Chhal	Birth	Natur	U	0.0	Newborn	—
22	DHAR	2024-11-17	Boro	Birth	Natur	U	0.05	Newborn	—
23	DHAR	2024-11-21	Chhal	Drowning	Natur	U	0.05	Newborn	—
24	DHAR	2025-01-21	Dharamjaigarh	Electrocution	Unnat	M	10	Sub-adult	Y
25	DHAR	2025-01-31	Dharamjaigarh	Old age	Natur	M	60	Adult	—
26	DHAR	2025-03-18	Chhal	Drowning	Natur	F	—	Calf	—
27	DHAR	2025-10-28	Chhal	Drowning	Natur	F	—	Calf	—
28	DHAR	2026-04-22	Lailunga	Disease	Natur	M	0.08	Newborn	—
29	DHAR	2026-05-08	Chhal	Disease	Natur	M	0.5	Calf	—
30	DHAR	2026-05-11	Chhal	Disease	Natur	M	0.46	Calf	—
31	DHAR	2026-05-24	Chhal	Drowning	Natur	M	1.75	Juvenile	—
32	RAIG	2022-06-12	Gharghoda	Heat stroke	Natur	U	0.12 5	Calf	—
33	RAIG	2022-11-20	Gharghoda	Electrocution	Unnat	F	11	Sub-adult	—
34	RAIG	2022-12-09	Gharghoda	Electrocution	Unnat	F	1	Juvenile	—
35	RAIG	2024-10-26	Tamnar	Electrocution	Unnat	F	2.5	Juvenile	—
36	RAIG	2024-10-26	Tamnar	Electrocution	Unnat	F	55	Adult	—
37	RAIG	2024-10-26	Tamnar	Electrocution	Unnat	M	65	Adult	—
38	RAIG	2024-12-31	Gharghoda	Drowning	Natur	M	2	Juvenile	—
39	RAIG	2025-01-14	Gharghoda	Drowning	Natur	F	0.2	Calf	—

40	RAIG	2025-01-22	Gharghoda	Drowning	Natur	F	2	Juvenile	—
41	RAIG	2025-05-12	Gharghoda	Injury	Natur	F	0.25	Calf	—
42	RAIG	2025-05-25	Gharghoda	Drowning	Natur	M	0.42	Calf	—
43	RAIG	2025-10-20	Tamnar	Electrocution	Unnat	M	20	Adult	—
44	RAIG	2025-11-25	Tamnar	Drowning	Natur	M	0.58	Calf	—
45	RAIG	2025-12-20	Raigarh	Drowning	Natur	M	0.5	Calf	—
46	RAIG	2026-01-28	Gharghoda	Injury	Natur	M	1	Juvenile	—
47	RAIG	2026-02-21	Tamnar	Weakness/emaciation	Natur	F	—	Calf	—
48	RAIG	2026-03-11	Gharghoda	Electrocution	Unnat	F	2.5	Juvenile	—
49	RAIG	2026-03-11	Gharghoda	Electrocution	Unnat	M	2.5	Juvenile	—
50	RAIG	2026-06-01	Kharsia	Drowning	Natur	M	0.3	Calf	—
51	RAIG	2026-06-16	Gharghoda	Train hit	Unnat	F	45	Adult	—

*Analytical report compiled from Dharamjaigarh & Raigarh Forest Division death registers · Workshop on Mortality Investigation of Asian Elephant ·
Raigarh, Chhattisgarh · June 2026*